
WORK

A Showcase of Three Key AI Projects



RAG Pipeline



Smart Compliance Tool



Advanced Research Agent

AGENDA

Today's Overview

01

RAG Pipeline & AI Chatbot

Architecture · Data Processing · Retrieval

02

Smart Compliance Tool

Architecture · User Features · AI Search

03

Deep Research & Search

Architecture · Real-Time UX · Performance

04

Other Projects

PUB Chatbot · DC Assistant T&C

05

Achievements & Skillset

Full-Stack AI Expertise

06

Q&A

Open Discussion

Three Distinct AI Systems

This presentation walks through the design, architecture, and key innovations of three high-impact AI-powered solutions — from foundational data pipelines to intelligent research agents.



Data Engineering

End-to-end pipelines from scraping to vector storage



AI Integration

LLMs, embeddings, hybrid search, and agent design



Product Thinking

Interactive UIs, compliance workflows, real-time UX

Foundational RAG Pipeline & AI Chatbot



The Challenge

Provide accurate, context-aware answers to complex queries from unstructured web and PDF documents — instantly and at scale.



The Solution

A fully automated, end-to-end Retrieval-Augmented Generation (RAG) pipeline — from web scraping and PDF ingestion to hybrid vector search and LLM response generation.



Acquisition

Scrape URA website + ingest PDFs automatically



Processing

Clean, structure, and enrich raw content into chunks



Storage

PostgreSQL + pgvector + Vercel Blob multi-tier system



Retrieval

Hybrid vector + keyword search for precise answers

Project 01 · System Architecture



Key Technical Highlights

Dual Scrape Modes

Daily incremental (new URLs) and weekly full refresh — automated by cron jobs, resumable via session management.

Advanced PDF Handling

PyMuPDF extracts embedded images and diagrams, linked contextually to text chunks for rich multi-modal retrieval.

Hybrid Search

80% vector similarity + 20% BM25 keyword search, with query expansion (e.g. GFA → Gross Floor Area) and cross-encoder reranking.

Monitoring Dashboard

Real-time UI tracks scraping health, content quality, and session status. Stuck sessions auto-cleaned after 24 hours.

Smart Compliance Operations Tool



The Challenge

Manual compliance checking is time-consuming, prone to inconsistency, and impossible to scale across large property portfolios.



The Solution

An AI-powered compliance tool that automates occupant identification, assesses properties against use regulations, and prioritises enforcement — at scale.



Automated Analysis

Assesses properties against approved-use regulations and identifies current occupants using AI.



Dual Input Modes

Bulk CSV uploads for large batches and single-entry forms for rapid ad-hoc checks.



Smart Prioritisation

Scores cases to intelligently rank enforcement actions by risk and urgency.



AI Search Engines

Explored SERP API (Google-backed) and Exa API (AI-native crawl) — Exa yielded stronger accuracy.

PROJECT 03 · Advanced Deep Research & Search Feature

Going Beyond Standard RAG

Standard RAG excels at retrieving facts from existing documents. But complex, real-world queries demand up-to-the-minute analysis from the live web. Project 3 delivers a multi-step deep research agent powered by Exa's Research API.



Web Search Mode

Real-time fact-checking via live web queries for quick answers.



Deep Research Mode

Multi-query parallel research with synthesised, citation-rich reports.

Research Architecture

1

Query Variation

Generates multiple phrasings to maximise coverage

2

Parallel Execution

All queries run simultaneously via Promise.all

3

Deduplication

Merges results, removes duplicate URLs and titles

4

Visualised Thought Process

ResearchSteps UI shows planning → searching → synthesis

5

Adaptive Timeouts

Extended waits for complex models (exa-research-pro)

Other Projects



PUB Chatbot

- Python-based scraper on Jupyter Notebook extracted PDFs and web content from the PUB website into Markdown.
- Implemented a custom chunking and embedding pipeline to load content into a vector database.
- Served as the knowledge base for a domain-specific RAG chatbot — a parallel architecture to Project 1.



DC Assistant — Terms & Conditions

- Drafted comprehensive T&C documentation for the DC Assistant product.
- General legal and privacy clauses benchmarked against PairChat and leading AI platforms.
- Usage and IP policies modelled on OpenAI, Gemini, and Anthropic's published policies.
- SLA framework crafted from multiple online references — identified as the section requiring the most refinement.

Comprehensive Skillset & Project Impact

01



RAG Pipeline

- End-to-End Data Pipelines
- Advanced RAG Systems
- Hybrid Information Retrieval
- Large-Scale Automation (Node.js / TypeScript)
- Database Architecture (PostgreSQL / pgvector)

02



Smart Compliance

- Interactive AI App Development
- Business Process Automation
- Rapid Prototyping (Python / Streamlit)
- Data Validation & Flexible API Integration

03



Research Agent

- Advanced AI Agent Design
- Exa API Integration
- Real-Time UX for Async Tasks
- Sophisticated Prompt Engineering
- Performance Optimisation

Full-stack AI expertise spanning backend data engineering, intelligent retrieval systems, and production-ready interactive frontends — across three distinct, high-impact solutions.

Q & A

Questions? Let's discuss.

Thank you for your time